Vector spherical harmonics and Mie scattering in optics and acoustics: conventions and explicit formulas

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This note collects the definitions and the explicit formulas for the scattering of electromagnetic and acoustic waves by a homogeneous sphere. It fixes one set of conventions for the spherical Bessel functions, the scalar and vector spherical harmonics, the plane-wave expansions, the multipole decomposition of a given field, and the electromagnetic and acoustic Mie problems. The electromagnetic part uses the dual-symmetric fields $\sqrt{\varepsilon}\mathbf{E}$ and $\sqrt{\mu}\mathbf{H}$ and the complex vector spherical harmonics of Ref. [1]. One sign convention for the T matrix covers both optics and acoustics.

I. CONVENTIONS

All fields are monochromatic with the time dependence $\mathrm{e}^{-\mathrm{i}\omega t}$. A real field is $\boldsymbol{\mathcal{A}}(\mathbf{r},t) = \mathrm{Re}[\mathbf{A}(\mathbf{r})\,\mathrm{e}^{-\mathrm{i}\omega t}]$. We use spherical coordinates (r,ϑ,φ) with the polar angle $\vartheta \in [0,\pi]$ and the unit vectors $\bar{\mathbf{r}}$, $\bar{\boldsymbol{\vartheta}}$, $\bar{\boldsymbol{\varphi}}$.

a. Electromagnetism. A homogeneous host has the absolute permittivity ε and permeability μ. A sphere of radius a has ε_p and μ_p. We use

$$k = \omega\sqrt{\varepsilon\mu}, \qquad k_p = \omega\sqrt{\varepsilon_p\mu_p} = mk, \qquad m = \sqrt{\bar{\varepsilon}\bar{\mu}}, \qquad \bar{\varepsilon} = \frac{\varepsilon_p}{\varepsilon}, \quad \bar{\mu} = \frac{\mu_p}{\mu}, \qquad x = ka. \tag{1}$$

The square roots take the principal branch, so $\mathrm{Im}\, k_p \geq 0$ for a passive particle. We write all electromagnetic formulas for the dual-symmetric pair $(\sqrt{\varepsilon}\mathbf{E}, \sqrt{\mu}\mathbf{H})$, as in Ref. [1]. With a dimensional amplitude $\mathcal{A}$, $[\mathcal{A}] = [\sqrt{\varepsilon}\mathbf{E}] = [\sqrt{\mu}\mathbf{H}]$, the normalized fields

$$\mathbf{e} = \frac{\sqrt{\varepsilon}\mathbf{E}}{\mathcal{A}}, \qquad \mathbf{h} = \frac{\sqrt{\mu}\mathbf{H}}{\mathcal{A}} \tag{2}$$

obey the source-free Maxwell equations

$$\boldsymbol{\nabla}\times\mathbf{e} = \mathrm{i}k\,\mathbf{h}, \qquad \boldsymbol{\nabla}\times\mathbf{h} = -\mathrm{i}k\,\mathbf{e}. \tag{3}$$

The time-averaged Poynting vector is $\mathbf{S} = \frac{1}{2}\,\mathrm{Re}(\mathbf{E}^*\times\mathbf{H}) = \frac{v}{2}|\mathcal{A}|^2\,\mathrm{Re}(\mathbf{e}^*\times\mathbf{h})$ with the phase velocity $v = 1/\sqrt{\varepsilon\mu}$. Equations (3) are invariant under the duality transformation

$$\mathbf{e} \to \mathbf{h}, \qquad \mathbf{h} \to -\mathbf{e}. \tag{4}$$

A field of helicity $\sigma = \pm 1$ obeys $\mathbf{h} = -\mathrm{i}\sigma\mathbf{e}$. The value $\sigma = +1$ is the left circular polarization (LCP) of Ref. [1], Chap. 4. The Riemann–Silberstein vectors $\mathbf{F}_\pm = (\mathbf{e} \pm \mathrm{i}\mathbf{h})/\sqrt{2}$ separate the two helicities: a field of helicity σ has $\mathbf{F}_{-\sigma} = 0$. The circular unit vectors are $\bar{\mathbf{e}}_\pm = (\bar{\mathbf{x}} \pm \mathrm{i}\bar{\mathbf{y}})/\sqrt{2}$. The cross sections need a lossless host, $\mathrm{Im}(\varepsilon\mu) = 0$, because in a lossy host the scattered field decays and its flux through a far sphere is not defined.

b. Acoustics. A fluid host has the density ρ and the compressibility β. The sphere has ρ_p and β_p. We use $k = \omega\sqrt{\rho\beta}$, $k_p = k\sqrt{\bar{\rho}\bar{\beta}}$, $\bar{\rho} = \rho_p/\rho$, $\bar{\beta} = \beta_p/\beta$, and the sound speed $c = 1/\sqrt{\rho\beta}$. The pressure p gives the velocity $\mathbf{v} = \boldsymbol{\nabla}p/(\mathrm{i}\omega\rho)$.

II. RADIAL FUNCTIONS

The radial functions $z_j^{(s)}(\rho)$, $s = 1,\ldots,4$, are

$$z_j^{(1)} = j_j, \qquad z_j^{(2)} = y_j, \qquad z_j^{(3)} = h_j^{(1)} = j_j + \mathrm{i}y_j, \qquad z_j^{(4)} = h_j^{(2)} = j_j - \mathrm{i}y_j. \tag{5}$$

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A field regular at the origin uses $s = 1$. An outgoing field uses $s = 3$. All four functions obey the same recurrences,

$$z'_j = z_{j-1} - \frac{j+1}{\rho} z_j = -z_{j+1} + \frac{j}{\rho} z_j, \tag{6}$$

$$z''_j = \frac{j\, z'_{j-1} - (j+1)\, z'_{j+1}}{2j+1} = -\frac{2}{\rho} z'_j - \left[1 - \frac{j(j+1)}{\rho^2}\right] z_j. \tag{7}$$

For $j = 0$ the term with z'_{j-1} has zero weight. The electric harmonic needs the combination

$$\zeta_j(\rho) = \frac{[\rho\, z_j(\rho)]'}{\rho} = \frac{z_j}{\rho} + z'_j. \tag{8}$$

For real ρ, z_j and ζ_j have no common zero, because ρz_j and its derivative cannot vanish together for a nonzero solution of a second-order equation. The Wronskians are

$$j_j y'_j - j'_j y_j = \frac{1}{\rho^2}, \qquad j_j h'_j - j'_j h_j = \frac{\mathrm{i}}{\rho^2}, \qquad \psi_n \xi'_n - \psi'_n \xi_n = \mathrm{i}, \tag{9}$$

with $h_j \equiv h_j^{(1)}$ and the Riccati–Bessel functions

$$\psi_n(\rho) = \rho\, j_n(\rho), \qquad \xi_n(\rho) = \rho\, h_n^{(1)}(\rho), \qquad \psi'_n = \psi_{n-1} - \frac{n}{\rho}\psi_n, \qquad \xi'_n = \xi_{n-1} - \frac{n}{\rho}\xi_n. \tag{10}$$

We also use the logarithmic derivative $D_n(z) = \psi'_n(z)/\psi_n(z)$. For $\rho \gg j^2$ the outgoing functions become

$$h_j(\rho) \simeq (-\mathrm{i})^{j+1} \frac{\mathrm{e}^{\mathrm{i}\rho}}{\rho}, \qquad \zeta_j^{(3)}(\rho) \simeq (-\mathrm{i})^j \frac{\mathrm{e}^{\mathrm{i}\rho}}{\rho}, \tag{11}$$

with corrections of order j^2/ρ.

III. SCALAR SPHERICAL HARMONICS AND LEGENDRE FUNCTIONS

We use the orthonormal harmonics with the Condon–Shortley phase $(-1)^m$ [1, 2],

$$Y_{mj}(\vartheta, \varphi) = \kappa_{mj}\, P_j^m(\cos\vartheta)\, \mathrm{e}^{\mathrm{i}m\varphi}, \qquad \kappa_{mj} = \sqrt{\frac{2j+1}{4\pi}\frac{(j-m)!}{(j+m)!}}, \qquad P_j^m(x) = (-1)^m (1-x^2)^{m/2} \frac{\mathrm{d}^m}{\mathrm{d}x^m} P_j(x), \tag{12}$$

so that $Y_{-m,j} = (-1)^m Y^*_{mj}$ and $\int_{4\pi} \mathrm{d}\Omega\, Y_{mj} Y^*_{m'j'} = \delta_{jj'}\delta_{mm'}$. The polar derivative is

$$\partial_\vartheta Y_{mj} = m \cot\vartheta\, Y_{mj} + \sqrt{(j-m)(j+m+1)}\, \mathrm{e}^{-\mathrm{i}\varphi}\, Y_{m+1,j}, \tag{13}$$

where the last term is absent for $m = j$.

a. Pole limits. The vector harmonics below contain $mY_{mj}/\sin\vartheta$. At the poles this ratio is finite and nonzero only for $m = \pm 1$. L'Hôpital's rule with $\sin\vartheta \simeq \vartheta$ or $\pi - \vartheta$ gives

$$\lim_{\vartheta\to 0} \frac{mY_{mj}}{\sin\vartheta} = m\, \partial_\vartheta Y_{mj}\big|_{\vartheta=0}, \qquad \lim_{\vartheta\to\pi} \frac{mY_{mj}}{\sin\vartheta} = -m\, \partial_\vartheta Y_{mj}\big|_{\vartheta=\pi}. \tag{14}$$

b. Angular functions of Bohren and Huffman. Bohren and Huffman [3] use the associated Legendre functions *without* the Condon–Shortley phase,

$$\mathcal{P}_n^m(x) = (-1)^m P_n^m(x) = (1-x^2)^{m/2} \frac{\mathrm{d}^m}{\mathrm{d}x^m} P_n(x). \tag{15}$$

With them we define, for $m \geq 0$,

$$\pi_n^m(\vartheta) = \frac{m\, \mathcal{P}_n^m(\cos\vartheta)}{\sin\vartheta}, \qquad \tau_n^m(\vartheta) = \frac{\mathrm{d}}{\mathrm{d}\vartheta} \mathcal{P}_n^m(\cos\vartheta), \qquad \pi_n \equiv \pi_n^1, \quad \tau_n \equiv \tau_n^1. \tag{16}$$

The dipole-type functions obey $\pi_0 = 0$, $\pi_1 = 1$, and

$$\pi_n = \frac{2n-1}{n-1}\, \mu\, \pi_{n-1} - \frac{n}{n-1}\, \pi_{n-2}, \qquad \tau_n = n\mu\, \pi_n - (n+1)\, \pi_{n-1}, \qquad \mu = \cos\vartheta. \tag{17}$$

Their pole values are

$$\pi_n(0) = \tau_n(0) = \frac{n(n+1)}{2}, \qquad \pi_n(\pi) = (-1)^{n+1} \frac{n(n+1)}{2}, \qquad \tau_n(\pi) = (-1)^n \frac{n(n+1)}{2}, \tag{18}$$

and $\pi_n^m = 0$ at both poles for $m \neq 1$. In general $\pi_n^m(0) = m\, \tau_n^m(0)$ and $\pi_n^m(\pi) = -m\, \tau_n^m(\pi)$, as in Eq. (14).

IV. VECTOR SPHERICAL HARMONICS

a. Angular harmonics. From Y_{mj} we build two tangential fields on the unit sphere,

$$\boldsymbol{\Psi}_{mj} = r\boldsymbol{\nabla}Y_{mj} = \bar{\boldsymbol{\vartheta}}\,\partial_\vartheta Y_{mj} + \bar{\boldsymbol{\varphi}}\,\frac{\mathrm{i}m}{\sin\vartheta}Y_{mj}, \qquad \boldsymbol{\Phi}_{mj} = \bar{\mathbf{r}}\times\boldsymbol{\Psi}_{mj} = -\bar{\boldsymbol{\vartheta}}\,\frac{\mathrm{i}m}{\sin\vartheta}Y_{mj} + \bar{\boldsymbol{\varphi}}\,\partial_\vartheta Y_{mj}. \tag{19}$$

With $Y_{mj}\bar{\mathbf{r}}$ they are orthogonal on the sphere,

$$\int_{4\pi}\mathrm{d}\Omega\;\boldsymbol{\Psi}_{mj}\cdot\boldsymbol{\Psi}^*_{m'j'} = \int_{4\pi}\mathrm{d}\Omega\;\boldsymbol{\Phi}_{mj}\cdot\boldsymbol{\Phi}^*_{m'j'} = j(j+1)\,\delta_{jj'}\delta_{mm'}, \qquad \int_{4\pi}\mathrm{d}\Omega\;\boldsymbol{\Psi}_{mj}\cdot\boldsymbol{\Phi}^*_{m'j'} = 0. \tag{20}$$

The harmonics of Jackson [4, Sec. 9.7], $\mathbf{X}_{lm} = \hat{\mathbf{L}}Y_{ml}/\sqrt{l(l+1)}$ with $\hat{\mathbf{L}} = -\mathrm{i}\mathbf{r}\times\boldsymbol{\nabla}$, are $\mathbf{X}_{jm} = -\mathrm{i}\,\boldsymbol{\Phi}_{mj}/\sqrt{j(j+1)}$. They vanish for $j = 0$.

b. Complex vector harmonics. Following Refs. [1, 2], we take the scalar solution $\psi_{mj} = z_j(kr)Y_{mj}(\vartheta,\varphi)$ and define the dimensionless harmonics

$$\mathbf{M}_{mj} = \boldsymbol{\nabla}\times(\mathbf{r}\,\psi_{mj}), \qquad \mathbf{N}_{mj} = \frac{1}{k}\boldsymbol{\nabla}\times\mathbf{M}_{mj}, \qquad \mathbf{L}_{mj} = \frac{1}{k}\boldsymbol{\nabla}\psi_{mj}. \tag{21}$$

With $\rho = kr$, Eqs. (8) and (19) give the compact forms

$$\mathbf{M}_{mj} = -z_j\,\boldsymbol{\Phi}_{mj}, \qquad \mathbf{N}_{mj} = j(j+1)\frac{z_j}{\rho}\,Y_{mj}\bar{\mathbf{r}} + \zeta_j\,\boldsymbol{\Psi}_{mj}, \qquad \mathbf{L}_{mj} = z'_j\,Y_{mj}\bar{\mathbf{r}} + \frac{z_j}{\rho}\,\boldsymbol{\Psi}_{mj}, \tag{22}$$

or, in components,

$$\begin{aligned}
\mathbf{M}_{mj} &= \bar{\boldsymbol{\vartheta}}\,\frac{\mathrm{i}m}{\sin\vartheta}z_jY_{mj} - \bar{\boldsymbol{\varphi}}\,z_j\,\partial_\vartheta Y_{mj},\\
\mathbf{N}_{mj} &= \bar{\mathbf{r}}\,j(j+1)\frac{z_j}{\rho}Y_{mj} + \bar{\boldsymbol{\vartheta}}\,\zeta_j\,\partial_\vartheta Y_{mj} + \bar{\boldsymbol{\varphi}}\,\frac{\mathrm{i}m}{\sin\vartheta}\zeta_jY_{mj},\\
\mathbf{L}_{mj} &= \bar{\mathbf{r}}\,z'_jY_{mj} + \bar{\boldsymbol{\vartheta}}\,\frac{z_j}{\rho}\,\partial_\vartheta Y_{mj} + \bar{\boldsymbol{\varphi}}\,\frac{\mathrm{i}m}{\sin\vartheta}\frac{z_j}{\rho}Y_{mj}.
\end{aligned} \tag{23}$$

The magnetic harmonic is also $\mathbf{M}_{mj} = -\mathrm{i}\hat{\mathbf{L}}\psi_{mj} = -\mathrm{i}\sqrt{j(j+1)}\,z_j\mathbf{X}_{jm}$. The harmonics obey

$$\boldsymbol{\nabla}\times\mathbf{M}_{mj} = k\mathbf{N}_{mj}, \qquad \boldsymbol{\nabla}\times\mathbf{N}_{mj} = k\mathbf{M}_{mj}, \qquad \boldsymbol{\nabla}\cdot\mathbf{M}_{mj} = \boldsymbol{\nabla}\cdot\mathbf{N}_{mj} = 0, \qquad \boldsymbol{\nabla}\times\mathbf{L}_{mj} = 0, \qquad \boldsymbol{\nabla}\cdot\mathbf{L}_{mj} = -k\psi_{mj}, \tag{24}$$

and, for real z_j, $\mathbf{W}_{-m,j} = (-1)^m\mathbf{W}^*_{mj}$ with $\mathbf{W} = \mathbf{M},\mathbf{N},\mathbf{L}$. On a sphere of radius r the nonzero integrals are [1, 5]

$$\begin{aligned}
&\int_{4\pi}\mathrm{d}\Omega\,|\mathbf{M}_{mj}|^2 = j(j+1)|z_j|^2, && \int_{4\pi}\mathrm{d}\Omega\,|\mathbf{N}_{mj}|^2 = \frac{j(j+1)}{2j+1}\left[(j+1)|z_{j-1}|^2 + j|z_{j+1}|^2\right],\\
&\int_{4\pi}\mathrm{d}\Omega\,|\mathbf{L}_{mj}|^2 = \frac{j|z_{j-1}|^2 + (j+1)|z_{j+1}|^2}{2j+1}, && \int_{4\pi}\mathrm{d}\Omega\,\mathbf{L}^*_{mj}\cdot\mathbf{N}_{mj} = \frac{j(j+1)}{2j+1}\left[|z_{j-1}|^2 - |z_{j+1}|^2\right].
\end{aligned} \tag{25}$$

At the origin the regular harmonics vanish, except $\mathbf{N}_{m1}(0) = \frac{2}{3}\boldsymbol{\nabla}(rY_{m1})$, which is a constant vector.

c. Vector components. A vector with spherical components $(A_r, A_\vartheta, A_\varphi)$ has the Cartesian components

$$\begin{pmatrix}A_x\\A_y\\A_z\end{pmatrix} = \begin{pmatrix}\sin\vartheta\cos\varphi & \cos\vartheta\cos\varphi & -\sin\varphi\\ \sin\vartheta\sin\varphi & \cos\vartheta\sin\varphi & \cos\varphi\\ \cos\vartheta & -\sin\vartheta & 0\end{pmatrix}\begin{pmatrix}A_r\\A_\vartheta\\A_\varphi\end{pmatrix}. \tag{26}$$

The columns are $\bar{\mathbf{r}}$, $\bar{\boldsymbol{\vartheta}}$, $\bar{\boldsymbol{\varphi}}$, and the inverse is the transpose. The angles of a point are $\vartheta = \mathrm{atan2}(\sqrt{x^2+y^2}, z)$ and $\varphi = \mathrm{atan2}(y,x)$.

V. PLANE WAVES

a. Scalar. A scalar plane wave with the unit wave vector $\bar{\mathbf{k}}$ is

$$\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} = 4\pi\sum_{j=0}^{\infty}\mathrm{i}^j\,j_j(kr)\sum_{m=-j}^{j}Y_{mj}(\bar{\mathbf{r}})\,Y^*_{mj}(\bar{\mathbf{k}}), \qquad \mathrm{e}^{\mathrm{i}kz} = \sum_{n=0}^{\infty}\mathrm{i}^n(2n+1)\,j_n(kr)\,P_n(\cos\vartheta). \tag{27}$$

b. Vector. A source-free field expands as [1]

$$\mathbf{e} = \sum_{mj} \left(A_{mj} \mathbf{N}_{mj} + B_{mj} \mathbf{M}_{mj} \right), \qquad \mathbf{h} = -\mathrm{i} \sum_{mj} \left(B_{mj} \mathbf{N}_{mj} + A_{mj} \mathbf{M}_{mj} \right), \tag{28}$$

with $\sum_{mj} \equiv \sum_{j=1}^{\infty} \sum_{m=-j}^{j}$. The electric amplitudes A_{mj} multiply $\mathbf{N}$ in $\mathbf{e}$. The magnetic amplitudes B_{mj} multiply $\mathbf{M}$ in $\mathbf{e}$. Equation (3) fixes $\mathbf{h}$, and the duality (4) acts as $(A_{mj}, B_{mj}) \to (-\mathrm{i}B_{mj}, -\mathrm{i}A_{mj})$. A plane wave $\mathbf{e} = \mathbf{e}_0 \, \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}$, $\mathbf{h} = (\bar{\mathbf{k}} \times \mathbf{e}_0) \, \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}$ with $\mathbf{e}_0 \cdot \bar{\mathbf{k}} = 0$ has $z_j = j_j$ and

$$A_{mj} = -\frac{4\pi \, \mathrm{i}^{j+1}}{j(j+1)} \, \mathbf{e}_0 \cdot \mathbf{\Psi}^*_{mj}(\bar{\mathbf{k}}), \qquad B_{mj} = -\frac{4\pi \, \mathrm{i}^{j}}{j(j+1)} \, \mathbf{e}_0 \cdot \mathbf{\Phi}^*_{mj}(\bar{\mathbf{k}}), \tag{29}$$

where $\mathbf{\Psi}_{mj}(\bar{\mathbf{k}})$ and $\mathbf{\Phi}_{mj}(\bar{\mathbf{k}})$ are Eq. (19) at the angles of $\bar{\mathbf{k}}$, written in Cartesian components with Eq. (26). For a general $\bar{\mathbf{k}}$, Eq. (29) contains all m.

For $\bar{\mathbf{k}} = \bar{\mathbf{z}}$ only $m = \pm 1$ survive. A circular wave $\mathbf{e}_0 = \bar{\mathbf{e}}_\sigma$ has

$$A_{mj} = \sigma \frac{\eta_j}{\sqrt{2}} \, \delta_{m\sigma}, \qquad B_{mj} = \frac{\eta_j}{\sqrt{2}} \, \delta_{m\sigma}, \qquad \eta_j = \mathrm{i}^{j+1} \sqrt{\frac{4\pi(2j+1)}{j(j+1)}}, \tag{30}$$

where η_j is β_j of Ref. [1]. A linear wave $\mathbf{e}_0 = \bar{\mathbf{x}}$ has $A_{\pm 1,j} = \pm \eta_j / 2$ and $B_{\pm 1,j} = \eta_j / 2$.

c. Helicity amplitudes. The combinations

$$C^\sigma_{mj} = \frac{A_{mj} + \sigma B_{mj}}{2}, \qquad \mathbf{e} = \sum_{\sigma=\pm} \sum_{mj} C^\sigma_{mj} \left(\mathbf{N}_{mj} + \sigma \mathbf{M}_{mj} \right), \qquad \mathbf{F}_\sigma = \sqrt{2} \sum_{mj} C^\sigma_{mj} \left(\mathbf{N}_{mj} + \sigma \mathbf{M}_{mj} \right) \tag{31}$$

split a field into its two helicities. The harmonic $\mathbf{N}_{mj} + \sigma \mathbf{M}_{mj}$ is an eigenfunction of the curl with the eigenvalue σk. The circular wave (30) has only $C^\sigma_{\sigma j} = \sigma \eta_j / \sqrt{2}$.

VI. DECOMPOSITION OF A GIVEN FIELD

a. Projections. Let a field on a sphere of radius r be $\mathbf{e} = \sum_{mj} (A_{mj} \mathbf{N}_{mj} + B_{mj} \mathbf{M}_{mj} + C_{mj} \mathbf{L}_{mj})$ with known z_j. The orthogonality (20) gives three projections,

$$\begin{aligned} e^Y_{mj} &\equiv \int_{4\pi} \mathrm{d}\Omega \, Y^*_{mj} \, (\bar{\mathbf{r}} \cdot \mathbf{e}) = j(j+1) \frac{z_j}{\rho} A_{mj} + z'_j C_{mj}, \\ e^\Psi_{mj} &\equiv \int_{4\pi} \mathrm{d}\Omega \, \mathbf{\Psi}^*_{mj} \cdot \mathbf{e} = j(j+1) \left[\zeta_j A_{mj} + \frac{z_j}{\rho} C_{mj} \right], \\ e^\Phi_{mj} &\equiv \int_{4\pi} \mathrm{d}\Omega \, \mathbf{\Phi}^*_{mj} \cdot \mathbf{e} = -j(j+1) \, z_j B_{mj}. \end{aligned} \tag{32}$$

The first two lines are a 2×2 system for A_{mj} and C_{mj}. For a source-free field $C_{mj} = 0$, and the magnetic field (28) adds the projections

$$h^Y_{mj} = -\mathrm{i}j(j+1) \frac{z_j}{\rho} B_{mj}, \qquad h^\Psi_{mj} = -\mathrm{i}j(j+1) \, \zeta_j B_{mj}, \qquad h^\Phi_{mj} = \mathrm{i}j(j+1) \, z_j A_{mj}. \tag{33}$$

The radial projections e^Y and h^Y alone give the projection formula of Ref. [1], Chap. 2:

$$A_{mj} = \frac{kr}{j(j+1) \, z_j(kr)} \int_{4\pi} \mathrm{d}\Omega \, Y^*_{mj} \, (\mathbf{e} \cdot \bar{\mathbf{r}}), \qquad B_{mj} = \frac{\mathrm{i}kr}{j(j+1) \, z_j(kr)} \int_{4\pi} \mathrm{d}\Omega \, Y^*_{mj} \, (\mathbf{h} \cdot \bar{\mathbf{r}}). \tag{34}$$

b. Zeros of the radial functions. Equation (34) divides by $z_j(kr)$. For an outgoing field $h_j(kr)$ has no real zeros. For a regular field Eq. (34) is undefined at the zeros of $j_j(kr)$. The tangential projections of $\mathbf{e}$ and $\mathbf{h}$ together determine the amplitudes at every radius, because z_j and ζ_j have no common zero. One such combination is

$$A_{mj} = \frac{\zeta^*_j \, e^\Psi_{mj} - \mathrm{i} z^*_j \, h^\Phi_{mj}}{j(j+1) \, (|z_j|^2 + |\zeta_j|^2)}, \qquad B_{mj} = \frac{-z^*_j \, e^\Phi_{mj} + \mathrm{i} \zeta^*_j \, h^\Psi_{mj}}{j(j+1) \, (|z_j|^2 + |\zeta_j|^2)}. \tag{35}$$

c. Scalar fields. A pressure field $p = \sum_{mj} C_{mj}\, z_j(kr)\, Y_{mj}$ gives

$$C_{mj} = \frac{1}{z_j(kr)} \int_{4\pi} \mathrm{d}\Omega\, Y^*_{mj}\, p, \qquad \int_{4\pi} \mathrm{d}\Omega\, Y^*_{mj}\, \rho c\, v_r = -\mathrm{i}\, C_{mj}\, z'_j(kr). \tag{36}$$

The two projections have no common zero, and they combine as in Eq. (35). A plane wave $p = p_0\, \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}$ has $C_{mj} = 4\pi\, \mathrm{i}^j\, Y^*_{mj}(\bar{\mathbf{k}})\, p_0$ with $z_j = j_j$.

VII. ELECTROMAGNETIC MIE PROBLEM

a. Fields. Outside the sphere the field is the incident part, with $z_j = j_j(kr)$ and amplitudes (A_{mj}, B_{mj}), plus the scattered part, with $z_j = h_j(kr)$ and amplitudes $(A^{\mathrm{sc}}_{mj}, B^{\mathrm{sc}}_{mj})$, both in the form (28). Inside, we normalize the fields with the particle parameters:

$$\frac{\sqrt{\varepsilon_p}\mathbf{E}}{\mathcal{A}} = \sum_{mj} \left(A^{\mathrm{in}}_{mj}\mathbf{N}_{mj} + B^{\mathrm{in}}_{mj}\mathbf{M}_{mj}\right), \qquad \frac{\sqrt{\mu_p}\mathbf{H}}{\mathcal{A}} = -\mathrm{i}\sum_{mj} \left(B^{\mathrm{in}}_{mj}\mathbf{N}_{mj} + A^{\mathrm{in}}_{mj}\mathbf{M}_{mj}\right), \tag{37}$$

with $z_j = j_j(k_p r)$. The tangential $\mathbf{E}$ and $\mathbf{H}$ are continuous at $r = a$.

b. Coefficients. The boundary conditions do not mix m, j, or the two types of harmonics. Thus the sphere is a diagonal T matrix [1, 6]. We define T as the outgoing amplitude divided by the regular amplitude of the same multipole:

$$A^{\mathrm{sc}}_{mj} = T^e_j A_{mj}, \qquad B^{\mathrm{sc}}_{mj} = T^m_j B_{mj}, \qquad A^{\mathrm{in}}_{mj} = \frac{\mathrm{i}m^2}{\mathcal{D}^a_j} A_{mj}, \qquad B^{\mathrm{in}}_{mj} = \frac{\mathrm{i}m^2}{\mathcal{D}^b_j} B_{mj}, \qquad T^e_n = -a_n, \quad T^m_n = -b_n. \tag{38}$$

Here a_n and b_n are the published Mie coefficients of Refs. [1, 3] [Appendix "Polarizabilities and Mie theory" of Ref. [1]],

$$a_n = \frac{\sqrt{\bar\varepsilon}\,\psi_n(mx)\psi'_n(x) - \sqrt{\bar\mu}\,\psi_n(x)\psi'_n(mx)}{\mathcal{D}^a_n}, \qquad b_n = \frac{\sqrt{\bar\mu}\,\psi_n(mx)\psi'_n(x) - \sqrt{\bar\varepsilon}\,\psi_n(x)\psi'_n(mx)}{\mathcal{D}^b_n}, \tag{39}$$

with the denominators

$$\mathcal{D}^a_n = \sqrt{\bar\varepsilon}\,\psi_n(mx)\xi'_n(x) - \sqrt{\bar\mu}\,\xi_n(x)\psi'_n(mx), \qquad \mathcal{D}^b_n = \sqrt{\bar\mu}\,\psi_n(mx)\xi'_n(x) - \sqrt{\bar\varepsilon}\,\xi_n(x)\psi'_n(mx). \tag{40}$$

The published coefficients carry the opposite sign to T: the scattered amplitude is $-a_n$ ($-b_n$) times the incident one. The coefficient b_n is a_n with $\bar\varepsilon \leftrightarrow \bar\mu$, and the same swap maps $\mathcal{D}^a_n \leftrightarrow \mathcal{D}^b_n$. The numerators of the internal amplitudes follow from the Wronskian (9). The coefficients a_n and b_n do not change under $m \to -m$, so the branch of m does not matter for them. Division of Eq. (39) by $\psi_n(mx)$ gives the equivalent form with $D_n = D_n(mx)$,

$$a_n = \frac{\sqrt{\bar\varepsilon}\,\psi'_n(x) - \sqrt{\bar\mu}\,D_n\,\psi_n(x)}{\sqrt{\bar\varepsilon}\,\xi'_n(x) - \sqrt{\bar\mu}\,D_n\,\xi_n(x)}, \qquad b_n = \frac{\sqrt{\bar\mu}\,\psi'_n(x) - \sqrt{\bar\varepsilon}\,D_n\,\psi_n(x)}{\sqrt{\bar\mu}\,\xi'_n(x) - \sqrt{\bar\varepsilon}\,D_n\,\xi_n(x)}. \tag{41}$$

c. Unitarity. The elements of the S matrix are $s^{e,m}_n = 1 + 2T^{e,m}_n$. A lossless sphere has $|s^{e,m}_n| = 1$, and a lossy sphere has $|s^{e,m}_n| < 1$.

d. Helicity and duality. In the helicity amplitudes (31) the T matrix (38) reads

$$C^{\sigma,\mathrm{sc}}_{mj} = \frac{1}{2}\left[(T^e_j + T^m_j)\, C^{\sigma}_{mj} + (T^e_j - T^m_j)\, C^{-\sigma}_{mj}\right]. \tag{42}$$

A dual sphere, $\bar\varepsilon = \bar\mu$, has $T^e_j = T^m_j$ and keeps the helicity of every multipole.

e. Resonances. The zeros of $\mathcal{D}^a_n$ in the complex plane of $\tilde x = \tilde k a$ are the poles of T^e_n, i.e. the electric (TM) modes of the sphere. The zeros of $\mathcal{D}^b_n$ are the poles of T^m_n, the magnetic (TE) modes. For $\bar\varepsilon = 16$, $\bar\mu = 1$ the magnetic dipole is at $\tilde x = 0.753782 - 0.024030\mathrm{i}$ and the electric dipole at $\tilde x = 1.052735 - 0.072355\mathrm{i}$. With $\mathrm{e}^{-\mathrm{i}\omega t}$ the poles have $\operatorname{Im}\tilde x < 0$, and the quality factor is $Q = \operatorname{Re}\tilde x/(2|\operatorname{Im}\tilde x|)$.

f. Limits. For $x \ll 1$ and $\bar\mu = 1$, the series of $a_1 = -T^e_1$ in Ref. [1] gives

$$T^e_1 \simeq \frac{2\mathrm{i}}{3}\frac{\bar\varepsilon - 1}{\bar\varepsilon + 2}x^3 + \mathrm{i}\frac{4 + \bar\varepsilon(\bar\varepsilon\bar\mu + \bar\varepsilon - 6)}{5(2+\bar\varepsilon)^2}x^5 - \frac{4}{9}\left(\frac{\bar\varepsilon - 1}{\bar\varepsilon + 2}\right)^2 x^6, \tag{43}$$

and T_1^m is the same with $\bar\varepsilon \leftrightarrow \bar\mu$. The dipole polarizabilities are $\alpha_e = -\mathrm{i}(6\pi/k^3)\,T_1^e$ and $\alpha_m = -\mathrm{i}(6\pi/k^3)\,T_1^m$, with the static limit $\alpha_e \to 4\pi a^3(\bar\varepsilon-1)/(\bar\varepsilon+2)$. A perfect electric conductor is the limit $|\bar\varepsilon|\to\infty$ with $\mathrm{Im}\,\sqrt{\bar\varepsilon}\to\infty$ at fixed $\bar\mu$. Then $D_n(mx)\to -\mathrm{i}$, and Eq. (41) gives

$$T_n^e = -\frac{\psi_n'(x)}{\xi_n'(x)}, \qquad T_n^m = -\frac{\psi_n(x)}{\xi_n(x)}, \tag{44}$$

with corrections of order $|\bar\varepsilon|^{-1/2}$.

VIII. FAR FIELD AND CROSS SECTIONS

a. Far field. With Eq. (11) the scattered field becomes a transverse spherical wave,

$$\mathbf{e}^{\mathrm{sc}} \simeq \frac{\mathrm{e}^{\mathrm{i}kr}}{kr}\,\mathbf{F}(\bar{\mathbf{r}}), \qquad \mathbf{F}(\bar{\mathbf{r}}) = \sum_{mj}(-\mathrm{i})^j\left[A^{\mathrm{sc}}_{mj}\boldsymbol{\Psi}_{mj}(\bar{\mathbf{r}}) + \mathrm{i}B^{\mathrm{sc}}_{mj}\boldsymbol{\Phi}_{mj}(\bar{\mathbf{r}})\right], \tag{45}$$

and the differential cross section of a plane wave $\mathbf{e}_0\mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}$ is $\mathrm{d}\sigma/\mathrm{d}\Omega = |\mathbf{F}|^2/(k^2|\mathbf{e}_0|^2)$.

b. Cross sections for any incident field. The orthogonality (20) and the flux of the cross term give [1, 2]

$$\sigma^{\mathrm{sc}} = \frac{1}{k^2|\mathbf{e}_0|^2}\sum_{mj} j(j+1)\left(|A^{\mathrm{sc}}_{mj}|^2 + |B^{\mathrm{sc}}_{mj}|^2\right), \qquad \sigma^{\mathrm{ext}} = -\frac{1}{k^2|\mathbf{e}_0|^2}\sum_{mj} j(j+1)\,\mathrm{Re}\left(A^*_{mj}A^{\mathrm{sc}}_{mj} + B^*_{mj}B^{\mathrm{sc}}_{mj}\right), \tag{46}$$

$$\sigma^{\mathrm{ext}} = \frac{4\pi}{k^2|\mathbf{e}_0|^2}\,\mathrm{Im}\left[\mathbf{e}_0^*\cdot\mathbf{F}(\bar{\mathbf{k}})\right], \qquad \sigma^{\mathrm{abs}} = \sigma^{\mathrm{ext}} - \sigma^{\mathrm{sc}}. \tag{47}$$

The second line is the optical theorem. For an incident beam that is not a plane wave, Eq. (46) keeps its form if we replace $|\mathbf{e}_0|^2$ by the chosen normalization of the incident intensity. For a sphere, $A^{\mathrm{sc}}_{mj} = T^e_j A_{mj}$ and $B^{\mathrm{sc}}_{mj} = T^m_j B_{mj}$ in Eq. (46).

c. Sphere in a plane wave. With Eqs. (38) and (29), Eq. (46) does not depend on the direction or the polarization of the wave:

$$\sigma^{\mathrm{sc}} = \frac{2\pi}{k^2}\sum_{n=1}^{\infty}(2n+1)\left(|T^e_n|^2 + |T^m_n|^2\right), \qquad \sigma^{\mathrm{ext}} = -\frac{2\pi}{k^2}\sum_{n=1}^{\infty}(2n+1)\,\mathrm{Re}(T^e_n + T^m_n), \tag{48}$$

$$\sigma^{\mathrm{abs}} = \frac{\pi}{2k^2}\sum_{n=1}^{\infty}(2n+1)\left(2 - |s^e_n|^2 - |s^m_n|^2\right). \tag{49}$$

The n-th electric (magnetic) contribution is the term with T^e_n (T^m_n). For example, the electric part of the absorption is $\sigma^{\mathrm{abs}}_{e,n} = -(2\pi/k^2)(2n+1)[\mathrm{Re}\,T^e_n + |T^e_n|^2]$. For $n=1$, Eq. (48) gives $\sigma^{\mathrm{ext}}_e = k\,\mathrm{Im}\,\alpha_e$ and $\sigma^{\mathrm{sc}}_e = (k^4/6\pi)|\alpha_e|^2$, as in Ref. [1].

d. Amplitude functions. For $\mathbf{e}_0 = \bar{\mathbf{x}}$ and $\bar{\mathbf{k}} = \bar{\mathbf{z}}$, the far field (45) is

$$F_\vartheta = \mathrm{i}\cos\varphi\; S_2(\vartheta), \qquad F_\varphi = -\mathrm{i}\sin\varphi\; S_1(\vartheta), \tag{50}$$

with the amplitude functions

$$S_1 = -\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}\left(T^e_n\pi_n + T^m_n\tau_n\right), \qquad S_2 = -\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}\left(T^e_n\tau_n + T^m_n\pi_n\right). \tag{51}$$

Equation (50) is $\mathbf{E}^{\mathrm{sc}}_{\parallel} = \mathrm{e}^{\mathrm{i}kr}S_2E_{\parallel}/(-\mathrm{i}kr)$ and $\mathbf{E}^{\mathrm{sc}}_{\perp} = \mathrm{e}^{\mathrm{i}kr}S_1E_{\perp}/(-\mathrm{i}kr)$ of Ref. [3]. The derived quantities are

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{|S_2|^2\cos^2\varphi + |S_1|^2\sin^2\varphi}{k^2}, \qquad \sigma^{\mathrm{sc}} = \frac{\pi}{k^2}\int_{-1}^{1}\mathrm{d}\mu\left(|S_1|^2 + |S_2|^2\right), \qquad \sigma^{\mathrm{ext}} = \frac{4\pi}{k^2}\,\mathrm{Re}\,S(0), \tag{52}$$

$$S(0) = S_1(0) = S_2(0) = -\frac{1}{2}\sum_n(2n+1)(T^e_n + T^m_n), \qquad \sigma^{\mathrm{back}} = \frac{4\pi}{k^2}|S_1(\pi)|^2 = \frac{\pi}{k^2}\left|\sum_n(2n+1)(-1)^n(T^e_n - T^m_n)\right|^2. \tag{53}$$

The asymmetry parameter $g = \langle \cos\vartheta \rangle$ and the radiation-pressure cross section are

$$g\,\sigma^{\rm sc} = \frac{4\pi}{k^2}\left[\sum_n \frac{n(n+2)}{n+1}\,\mathrm{Re}\left(T_n^e T_{n+1}^{e*} + T_n^m T_{n+1}^{m*}\right) + \sum_n \frac{2n+1}{n(n+1)}\,\mathrm{Re}\left(T_n^e T_n^{m*}\right)\right], \tag{54}$$

$$\sigma^{\rm pr} = \sigma^{\rm ext} - g\,\sigma^{\rm sc}. \tag{55}$$

Equations (51)–(54) are the formulas of Ref. [3, Ch. 4] with $a_n = -T_n^e$ and $b_n = -T_n^m$. The normalization $\sigma/(\pi a^2)$ gives the efficiencies Q of Ref. [3].

IX. REAL HARMONICS OF BOHREN AND HUFFMAN

This section maps the real basis of Ref. [3] to the complex form above.

a. Harmonics. For $m \geq 0$ the generating functions are

$$\psi_{emn} = \cos(m\varphi)\,\mathcal{P}_n^m(\cos\vartheta)\,z_n(kr), \qquad \psi_{omn} = \sin(m\varphi)\,\mathcal{P}_n^m(\cos\vartheta)\,z_n(kr), \tag{56}$$

and we define $\mathbf{M}_{\substack{e\\o}mn} = \boldsymbol{\nabla} \times \left(\mathbf{r}\psi_{\substack{e\\o}mn}\right)$ and $\mathbf{N}_{\substack{e\\o}mn} = k^{-1}\boldsymbol{\nabla} \times \mathbf{M}_{\substack{e\\o}mn}$. In components, with Eq. (16),

$$\begin{aligned}
\mathbf{M}_{emn} &= -\bar{\boldsymbol{\vartheta}}\sin(m\varphi)\,\pi_n^m z_n - \bar{\boldsymbol{\varphi}}\cos(m\varphi)\,\tau_n^m z_n,\\
\mathbf{M}_{omn} &= \bar{\boldsymbol{\vartheta}}\cos(m\varphi)\,\pi_n^m z_n - \bar{\boldsymbol{\varphi}}\sin(m\varphi)\,\tau_n^m z_n,\\
\mathbf{N}_{emn} &= \bar{\mathbf{r}}\,n(n+1)\frac{z_n}{\rho}\cos(m\varphi)\mathcal{P}_n^m + \left[\bar{\boldsymbol{\vartheta}}\cos(m\varphi)\tau_n^m - \bar{\boldsymbol{\varphi}}\sin(m\varphi)\pi_n^m\right]\zeta_n,\\
\mathbf{N}_{omn} &= \bar{\mathbf{r}}\,n(n+1)\frac{z_n}{\rho}\sin(m\varphi)\mathcal{P}_n^m + \left[\bar{\boldsymbol{\vartheta}}\sin(m\varphi)\tau_n^m + \bar{\boldsymbol{\varphi}}\cos(m\varphi)\pi_n^m\right]\zeta_n.
\end{aligned} \tag{57}$$

The sign of $\mathcal{P}_n^m$ matters. With the Condon–Shortley P_n^m instead of Eq. (15), every harmonic with odd m changes sign, and the plane-wave series (60) gives $-\mathbf{e}$.

b. Map to the complex harmonics. For $m \geq 0$ and $\mathbf{W} = \mathbf{M}, \mathbf{N}$, with any z_n,

$$\mathbf{W}_{mj} = (-1)^m \kappa_{mj}\left(\mathbf{W}_{emj} + \mathrm{i}\mathbf{W}_{omj}\right), \qquad \mathbf{W}_{-m,j} = \kappa_{mj}\left(\mathbf{W}_{emj} - \mathrm{i}\mathbf{W}_{omj}\right), \tag{58}$$

$$\mathbf{W}_{emj} = \frac{(-1)^m\mathbf{W}_{mj} + \mathbf{W}_{-m,j}}{2\kappa_{mj}}, \qquad \mathbf{W}_{omj} = \frac{(-1)^m\mathbf{W}_{mj} - \mathbf{W}_{-m,j}}{2\mathrm{i}\kappa_{mj}}, \qquad \mathbf{W}_{e0j} = \frac{\mathbf{W}_{0j}}{\kappa_{0j}}, \quad \mathbf{W}_{o0j} = 0. \tag{59}$$

Equation (25) of the Supplemental Material of Ref. [2] has no factor $(-1)^m$, because there the real harmonics are built on P_n^m with the Condon–Shortley phase. With the book functions $\mathcal{P}_n^m$ of Eq. (15), the map is Eq. (58), where κ_{mj} is β_{mj} of Ref. [2].

c. Fields of an x-polarized wave. For $\mathbf{e}_0 = \bar{\mathbf{x}}$, $\bar{\mathbf{k}} = \bar{\mathbf{z}}$, and $E_n = \mathrm{i}^n(2n+1)/[n(n+1)]$, Ref. [3, Ch. 4] gives, in our normalization,

$$\mathbf{e}^{\rm inc} = \sum_n E_n\left(\mathbf{M}_{o1n}^{(1)} - \mathrm{i}\mathbf{N}_{e1n}^{(1)}\right), \qquad \mathbf{h}^{\rm inc} = -\sum_n E_n\left(\mathbf{M}_{e1n}^{(1)} + \mathrm{i}\mathbf{N}_{o1n}^{(1)}\right), \tag{60}$$

$$\mathbf{e}^{\rm sc} = -\sum_n E_n\left(\mathrm{i}T_n^e\mathbf{N}_{e1n}^{(3)} - T_n^m\mathbf{M}_{o1n}^{(3)}\right), \qquad \mathbf{h}^{\rm sc} = -\sum_n E_n\left(\mathrm{i}T_n^m\mathbf{N}_{o1n}^{(3)} + T_n^e\mathbf{M}_{e1n}^{(3)}\right), \tag{61}$$

$$\frac{\sqrt{\varepsilon_p}\mathbf{E}}{\mathcal{A}} = \sqrt{\bar{\varepsilon}}\sum_n E_n\left(c_n\mathbf{M}_{o1n}^{(1)} - \mathrm{i}d_n\mathbf{N}_{e1n}^{(1)}\right), \qquad \frac{\sqrt{\mu_p}\mathbf{H}}{\mathcal{A}} = -\sqrt{\bar{\varepsilon}}\sum_n E_n\left(d_n\mathbf{M}_{e1n}^{(1)} + \mathrm{i}c_n\mathbf{N}_{o1n}^{(1)}\right), \tag{62}$$

where the superscript is s of Eq. (5), the internal harmonics have $\rho = k_p r$, and $\mathcal{A} = \sqrt{\varepsilon}E_0$. The internal coefficients of Ref. [3] are

$$c_n = \frac{\mathrm{i}\bar{\mu}\sqrt{\bar{\varepsilon}}}{\mathcal{D}_n^b}, \qquad d_n = \frac{\mathrm{i}\bar{\mu}\sqrt{\bar{\varepsilon}}}{\mathcal{D}_n^a}, \qquad \sqrt{\bar{\varepsilon}}\,c_n = \frac{\mathrm{i}m^2}{\mathcal{D}_n^b}, \qquad \sqrt{\bar{\varepsilon}}\,d_n = \frac{\mathrm{i}m^2}{\mathcal{D}_n^a}, \tag{63}$$

so that Eq. (62) matches Eq. (38).

X. ACOUSTIC MIE PROBLEM

a. Fields. An incident plane wave along z and the scattered and internal pressures are [1, 7]

$$p^{\rm inc}=\sum_{n=0}^{\infty}p_n j_n(kr)P_n(\cos\vartheta),\qquad p^{\rm sc}=\sum_{n=0}^{\infty}p_n a_n h_n(kr)P_n(\cos\vartheta),\qquad p^{\rm in}=\sum_{n=0}^{\infty}p_n c_n j_n(k_p r)P_n(\cos\vartheta),\tag{64}$$

with $p_n=p_0\mathrm{i}^n(2n+1)$. The pressure and $\rho^{-1}\partial_r p$ are continuous at $r=a$. They give [1, 7]

$$a_n=\frac{\sqrt{\bar\beta}\,j_n'(k_pa)\,j_n(ka)-\sqrt{\bar\rho}\,j_n(k_pa)\,j_n'(ka)}{\mathcal{D}_n},\qquad c_n=\frac{\mathrm{i}\sqrt{\bar\rho}}{(ka)^2\,\mathcal{D}_n},$$
$$\mathcal{D}_n=\sqrt{\bar\rho}\,j_n(k_pa)\,h_n'(ka)-\sqrt{\bar\beta}\,j_n'(k_pa)\,h_n(ka).\tag{65}$$

With the definition of T in Sec. VII, the published acoustic coefficient already has the sign of T: $T_n=+a_n$. For any incidence, $p^{\rm inc}=p_0\sum_{mj}C_{mj}j_j(kr)Y_{mj}$ with C_{mj} from Sec. VI, the scattered amplitudes are $C^{\rm sc}_{mj}=T_jC_{mj}$. As in optics, $s_n=1+2T_n$ has $|s_n|=1$ for a lossless sphere and $|s_n|<1$ with loss.

b. Limits. The relative impedance is $\bar Z=\sqrt{\bar\rho/\bar\beta}=\rho_pc_p/(\rho c)$. At fixed k_pa,

$$\bar Z\to\infty\text{ (rigid)}:\quad T_n=-\frac{j_n'(ka)}{h_n'(ka)},\qquad \bar Z\to0\text{ (sound soft)}:\quad T_n=-\frac{j_n(ka)}{h_n(ka)}.\tag{66}$$

With one sign of T, the rigid sphere has the structure of T^e_n of a perfect conductor (44), and the soft sphere that of T^m_n. For $ka\ll1$ [1],

$$T_0=a_0\simeq\frac{\mathrm{i}}{3}(\bar\beta-1)(ka)^3+\frac{\mathrm{i}}{45}\left[\bar\beta^2(\bar\rho+5)-15\bar\beta+9\right](ka)^5-\frac{1}{9}(\bar\beta-1)^2(ka)^6,$$
$$T_1=a_1\simeq\frac{\mathrm{i}}{3}\frac{\bar\rho-1}{2\bar\rho+1}(ka)^3+\frac{\mathrm{i}}{5}\frac{\bar\rho^2(\bar\beta-1)-\bar\rho+1}{(2\bar\rho+1)^2}(ka)^5-\frac{1}{9}\left(\frac{\bar\rho-1}{2\bar\rho+1}\right)^2(ka)^6.\tag{67}$$

c. Far field and cross sections. The scattered pressure becomes $p^{\rm sc}\simeq p_0f(\vartheta)\mathrm{e}^{\mathrm{i}kr}/r$ with

$$f(\vartheta)=-\frac{\mathrm{i}}{k}\sum_n(2n+1)\,T_nP_n(\cos\vartheta),\qquad f(\bar{\mathbf{r}})=\frac{1}{k}\sum_{mj}(-\mathrm{i})^{j+1}C^{\rm sc}_{mj}Y_{mj}(\bar{\mathbf{r}})\text{ (any incidence)}.\tag{68}$$

The cross sections are

$$\sigma^{\rm sc}=\int\mathrm{d}\Omega\,|f|^2=\frac{4\pi}{k^2}\sum_n(2n+1)|T_n|^2,\qquad \sigma^{\rm ext}=\frac{4\pi}{k}\,\mathrm{Im}\,f(0)=-\frac{4\pi}{k^2}\sum_n(2n+1)\,\mathrm{Re}\,T_n,$$
$$\sigma^{\rm abs}=\sigma^{\rm ext}-\sigma^{\rm sc}=\frac{\pi}{k^2}\sum_n(2n+1)\left(1-|s_n|^2\right),\tag{69}$$

$$\sigma^{\rm pr}=\sigma^{\rm ext}-\int\mathrm{d}\Omega\,\cos\vartheta\,|f|^2=-\frac{4\pi}{k^2}\sum_n\left[(2n+1)\,\mathrm{Re}\,T_n+2(n+1)\,\mathrm{Re}\left(T_nT^*_{n+1}\right)\right].\tag{70}$$

The radiation force of the plane wave is $F_z=\beta|p_0|^2\sigma^{\rm pr}/2$ [1, 8]. For any incidence, $\sigma^{\rm sc}=k^{-2}\sum_{mj}|C^{\rm sc}_{mj}|^2$ and $\sigma^{\rm ext}=-k^{-2}\sum_{mj}\mathrm{Re}(C^*_{mj}C^{\rm sc}_{mj})$, with $|p_0|=1$. For $ka\ll1$, Eq. (67) gives $\sigma^{\rm sc}\simeq4\pi k^4a^6\left(|f_1|^2/9+|f_2|^2/12\right)$ with $f_1=1-\bar\beta$ and $f_2=2(\bar\rho-1)/(2\bar\rho+1)$.

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